NEO Factory: from 24-minute to 2.4-minute takt

To: Vikram Kothari, VP Manufacturing & Hardware, 1X · **From:** Tristan Massicotte, Industrial Engineering, University of Toronto · October 2026

Going from 10,000 NEOs a year to 100,000 changes the line from one robot every 24 minutes to one every 2.4. I spent 2026 building the planning systems of a mid-size manufacturer in Québec: an MRP that cut order processing time by 90%, and a production board that tracks inventory along the assembly line. This memo is how I would start working on your ramp.

It uses public information only. Every number I assumed is labelled as such, and the spreadsheet behind it lets you replace my assumptions with real data.

1. The takt math

At 100,000 robots a year on two shifts, the line must finish a robot every 2.4 minutes, and the motor lines must finish a motor every 1.9 seconds.

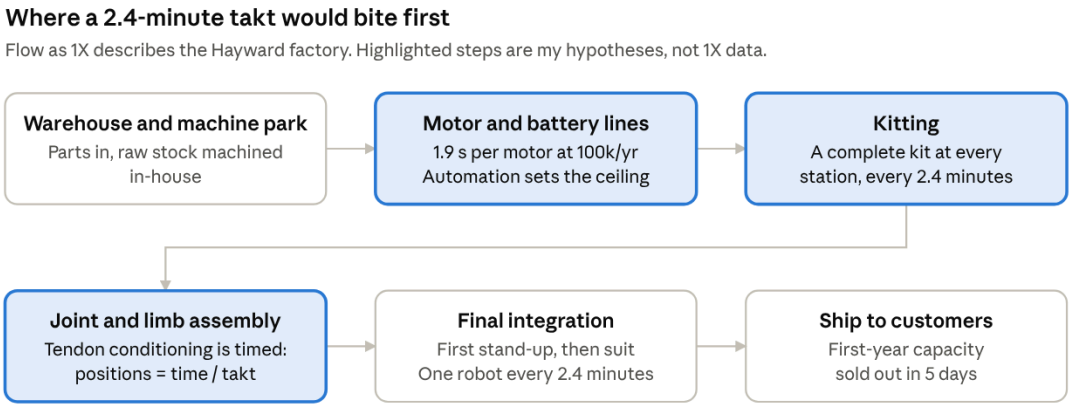
Annual volume (robots)	Robot takt (min)	Motors per year	Motor takt (sec)
10,000	24.0	750,000	19.2
25,000	9.6	1,875,000	7.7
50,000	4.8	3,750,000	3.8
100,000	2.4	7,500,000	1.9

Assumptions: 250 days × 2 shifts × 8 h = 4,000 h/yr at 100% availability (real takt is tighter). 75 motors per robot, from NEO's published 75 degrees of freedom at one motor per joint.

Reports differ on whether 100,000 is a yearly rate or a total by end-2027. Either way, most of it is built in 2027, so 2027 runs close to this pace. Whatever the real work content per robot, the same design needs about 10 times the stations, or stations 10 times faster.

2. Line walk: where 2.4 minutes bites first

Three steps worry me most, because each one grows by adding parallel capacity, not by working faster.



- **Motor and battery lines.** At 100,000 robots a year, the 17,000 motors built at Hayward so far would be about 9 hours of output. The cycle time of the automated cells sets the ceiling.
- **Kitting.** Single-piece flow needs a complete kit at every station, every 2.4 minutes. Kitting becomes a line of its own, with its own takt.
- **Tendon conditioning.** Any timed step needs positions = step time ÷ takt. For example, a 60-minute step needs 3 positions at 24 minutes and 25 positions at 2.4.

3. Materials: single-piece flow makes clear-to-build the number to watch

With pre-kitted modules at every station, one missing part stops the line, not a department. Three questions I would bring on day one:

1. **What is the clear-to-build today?** For each of the five commodity areas in your planning job posting, divide the complete kits on hand by the kits per robot. The lowest result is how many robots you can start. Hand and arm mechanisms count twice per robot, so that is where I would look first.
2. **How are hands counted?** NEO needs 2 hands. 10,000 robots need 20,000. The public 2026 target is 10,000 hands. If that counts single hands, it covers about 5,000 robots. It may well be counted in pairs, so I would ask before assuming a gap.
3. **What does the new hand do to stock already ordered?** The 25-DoF hand announced in July is going on every NEO. Parts for the old version still on hand or on order at the cut-in date become excess and obsolete stock. The levers are the cut-in week and each supplier's cancellation window. The spreadsheet shows the trade-off week by week.

4. My first 12 weeks, and my ask

1. **Weeks 1–2: work the line.** A shift at each station in joint and limb assembly and final integration. Time-study real cycles and learn the factory OS.
2. **Weeks 3–6: own one shortage loop.** Pick one module, for example forearm and hand. Log every line stop caused by an incomplete kit, find the root causes, and publish a daily clear-to-build report.
3. **Weeks 7–10: get that module ready for the higher rate.** Line balance at the target takt, the station count, and the kit delivery rhythm for when San Carlos comes up.
4. **Weeks 11–12: hand it off.** Standard work written down, and the model running on real numbers.

> **My ask:** a summer 2027 internship in production control or planning, on the floor in Hayward or San Carlos. As a Canadian student I would come on a J-1 intern visa. I handle the paperwork; 1X signs a training plan.

Scan for the spreadsheet model, my resume and what I built at Technimount.
tristan-massicotte.github.io/neo-ramp

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Sources. 1X, "NEO Factory | Building Your NEO," 1x.tech/discover/neo-factory, Apr 30 2026 (capacity, 100k+ target, single-piece flow, pre-kitting, motor lines, 17,000 motors built) · 1X factory opening release, GlobeNewswire, Apr 30 2026 · humanoid.guide NEO spec sheet (75 degrees of freedom) · humanoid.guide, "NEO Gets New Hands," Jul 2026 (25-DoF hand, 10,000 hands targeted in 2026) · 1X job posting, Supply Chain Planner, Advanced Robotics Commodities (five commodity areas) · Silicon Report summary of Bloomberg's factory report (100,000 total by end-2027).